Mini Technical Report

Mechanistic Analysis of Shortcut Learning in a Toy Induction Transformer

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https://github.com/hamzamooraj99/tiny-transformer-mech-interp

1. INTRODUCTION

Transformers trained on synthetic autoregressive tasks can attain low loss via mechanisms that differ from the intended algorithmic computation [2, 5]. In periodic induction-style sequence tasks, a canonical solution is an *induction head* [4] that attends to a prior occurrence at a fixed offset, yielding a diagonal attention structure. However, whether such circuits emerge depends on architectural inductive biases and on how strongly the task constrains alternative shortcut strategies.

This project studies a minimal decoder-only transformer trained on a synthetic periodic induction dataset. The aim is not performance optimisation, but mechanistic characterisation: induction-style attention structure and diagnose shortcut mechanisms are tested using targeted attention probes.

2. BASELINE TASK, MODEL, & PERFORMANCE

2.1. Dataset

A synthetic periodic induction dataset is constructed with:

- Vocabulary size $V = 20$
- Pattern length $P = 8$
- Sequence length $T = 32$
- Batch size $B = 64$

For each sequence, a base pattern of length P is sampled uniformly from $\{0, \dots, V-1\}$ and repeated to length T. The training objective is standard autoregressive next-token prediction on the full sequence, with no corruption. Fresh patterns are sampled per batch to avoid memorisation of a fixed dataset. For run-to-run reproducibility, a global random seed is set once at initialisation (`seed` $= 42$), without re-seeding per batch.

2.2. Model and Optimisation

The project surrounds a small causal transformer with:

- $n_{\text{params}} = 104,704$
- $n_{\text{layers}} = 2$
- $n_{\text{heads}} = 4$
- $d_{\text{model}} = 64$
- MLP hidden size $= 256$
- Learned token embeddings and learned positional embeddings

Optimisation uses cross-entropy loss with gradient clipping, learning rate 1×10^{-3}, and 2000 update steps. The final training loss under this baseline setup is 0.6807.

2.3. Quantitative Performance

Evaluation over 50 freshly generated batches is done from the same data-generating process (no corruption). Metrics are computed under teacher forcing and partitioned into: (i) *early* positions (before the first full pattern instance has been observed), and (ii) the *induction region* (positions $\geq P$), where the sequence becomes deterministic under periodic repetition.

The model achieves:

- Full-sequence: `eval/loss` $= 0.678$, `eval/acc` $= 0.7853$
- Induction region ($\geq P$): `eval/induction/loss` $= 4.676 \times 10^{-4}$, `eval/induction/acc` $= 1.0$

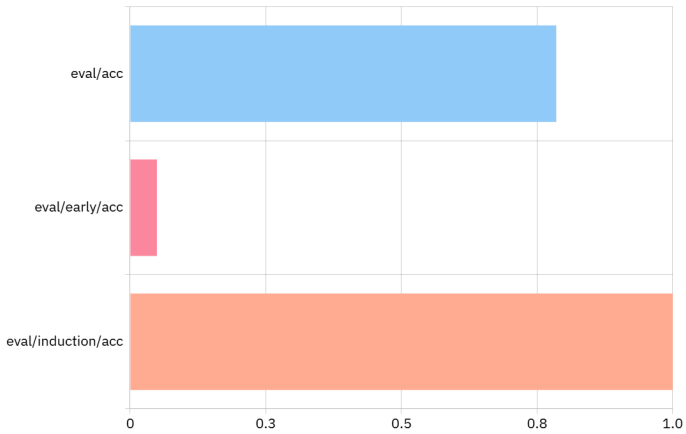

Figure 1. Accuracy scores for different regions of the sequence. Blue shows full-sequence accuracy, pink shows early region accuracy, and orange shows induction region accuracy

- Early region ($< P$): loss $= 3.001$, accuracy $= 0.0493$

Early accuracy is near the random-guess baseline ($1/V = 0.05$), consistent with the initial pattern tokens being information-theoretically unpredictable prior to observing the full pattern. In contrast, perfect induction-region accuracy indicates the model has learned a reliable predictive strategy for the repeated portion of the sequence. Subsequent sections test whether this strategy corresponds to induction-style attention circuits or shortcut mechanisms enabled by positional structure.

3. BASELINE MECHANISTIC ANALYSIS

Although the model achieves perfect accuracy within the induction region (Section 2.3), it remains unclear whether this behaviour arises from the canonical *induction head* mechanism, commonly observed in transformers trained on repeated-sequence tasks [1, 4]. To investigate the strategy learned by the model, targeted attention probes were applied to the trained network [3].

3.1. Offset-Diagonal Attention

The defining signature of an induction head is strong attention to the previous occurrence of a token. This corresponds to the offset diagonal $j = i - P$ where P is the pattern length. To test for this behaviour, the mean attention weight assigned to this diagonal was computed for each attention head across ten independently generated probe sequences.

As shown in Figure 2, heads in the first layer exhibit only weak preference for the induction diagonal, with mean attention values between 0.026 and 0.058. These values are only slightly above the uniform-attention baseline (≈ 0.032 for $T = 31$). In contrast, second-layer heads show negligible diagonal attention, with values below 0.008 across all heads. This indicates that the model does not rely on a strong induction-style copying mechanism.

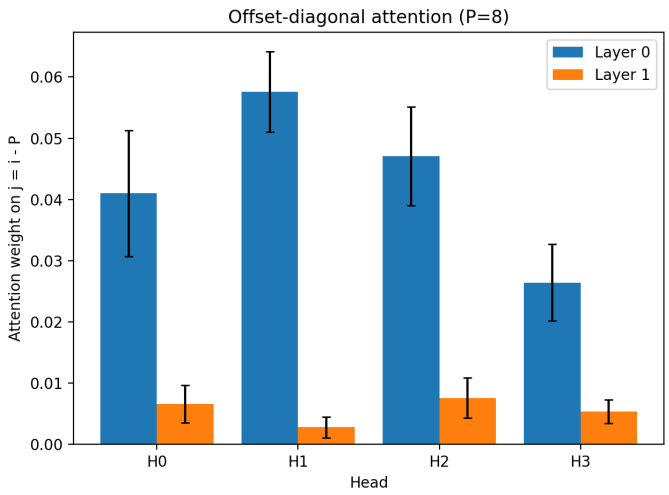

Figure 2. Mean ± standard deviation attention weight assigned to the offset diagonal $j = i - P$ across attention heads.

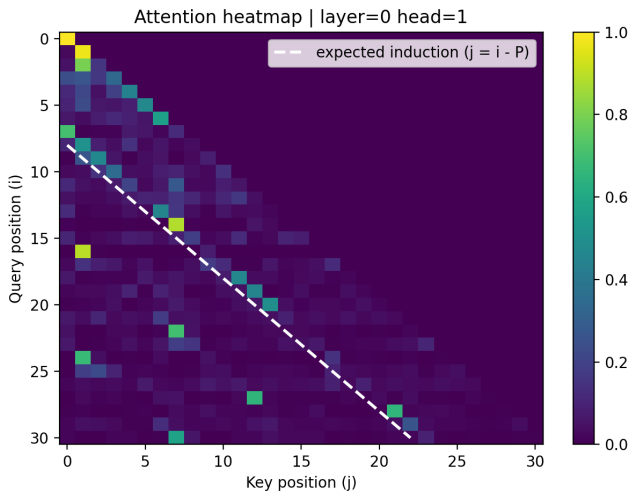

Figure 3. Attention heatmap for Layer 0 Head 1 on a representative probe sequence. The dashed line indicates the expected induction diagonal ($j = i - P$).

3.2. Attention Structure

To qualitatively examine the learned attention patterns, Figure 3 visualises the attention map of the head with the highest diagonal score (Layer 0, Head 1).

If the model had learned a canonical induction head, attention would concentrate along the diagonal corresponding to $j = i - P$. Instead, attention is distributed across multiple prior tokens and appears slightly offset from the expected induction diagonal. This suggests that the head exhibits only a weak near-diagonal preference rather than a clean induction-style copying mechanism. Attention mass is distributed across multiple earlier tokens rather than focused on a single previous occurrence.

3.3. Interpretation

These results suggest that the model does not implement a clean induction circuit. While some early-layer heads exhibit a mild preference for the induction offset, the signal is weak and does not propagate to later layers. Nevertheless, the model achieves perfect predictive performance within the repeated portion of the sequence. This indicates that the task can be solved through alternative attention strategies that do not rely on the canonical induction mechanism.

This discrepancy between behavioural performance and internal structure motivates intervention.

4. INTERVENTION 1 – POSITIONAL ABLATION

4.1. Intervention Design

The baseline analysis suggested that the model does not implement a canonical induction head. This raises the possibility that the learned strategy depends on positional information to identify the previous occurrence of the pattern.

To test this hypothesis, positional embeddings were ablated at evaluation time while keeping all model weights fixed. The model was then re-evaluated on the same dataset and the probes described in Section 3 were re-computed.

4.2. Results

Removing positional embeddings caused a dramatic degradation in performance. Full-sequence accuracy dropped to 0.094, close to the random baseline of $1/V = 0.05$. Accuracy within the induction region also collapsed to 0.106, indicating that the model can no longer exploit the repeated structure of the sequence.

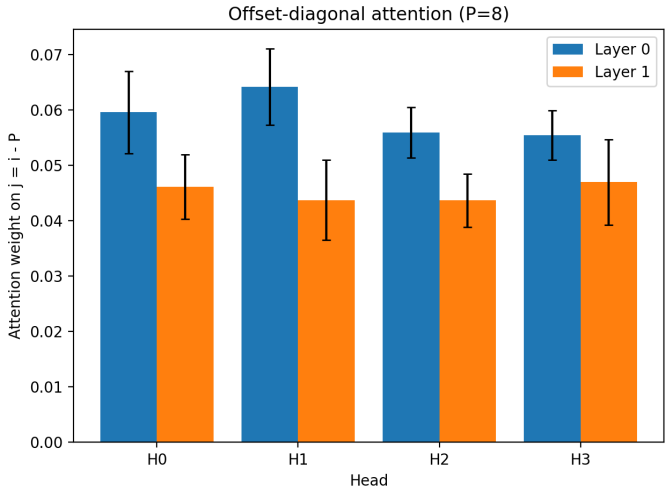

Figure 4. Offset-diagonal attention after positional ablation.

Despite the behavioural collapse, the offset-diagonal probe shows that several heads still assign slightly elevated attention to the expected induction offset $j = i - P$ (Figure 4). However, these values remain small and do not indicate a strong induction-style circuit.

Attention maps exhibit smooth recency-like gradients rather than sharp induction-style patterns (Figure 5).

4.3. Mechanistic Analysis

These results indicate that the baseline strategy depends critically on positional information. Once positional embeddings are removed, the model loses the ability to exploit the repeated structure of the sequence and its predictions revert to near-random behaviour.

The persistence of weak diagonal attention further suggests that attention patterns alone do not constitute a functional induction mechanism. The model may still attend near the expected offset, but without positional alignment the attended representations no longer support accurate next-token prediction. Thus confirming that positional information is essential for the strategy learned by the model.

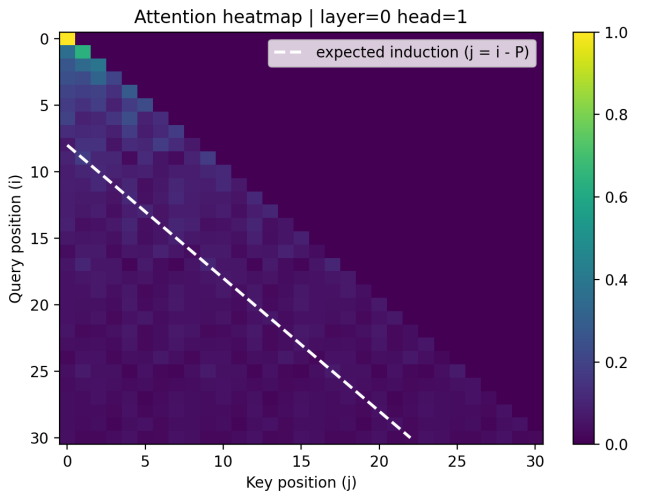

Figure 5. Attention heatmap for Layer 0 Head 1 after positional ablation.

5. INTERVENTION 2 – PHASE ROTATION

5.1. Intervention Design

To test whether the learned strategy depends on a fixed alignment between the repeated pattern and absolute sequence positions, the evaluation sequences were subjected to a global phase rotation. Specifically, the generated sequence was cyclically shifted along the sequence dimension using a tensor roll operation. This transformation preserves the periodic structure of the sequence while altering the phase at which the pattern appears relative to the start of the sequence.

5.2. Results

Evaluation under phase rotation produced performance nearly identical to the baseline model. Full-sequence accuracy and induction-region accuracy remained unchanged, and the attention probes revealed no substantial difference in offset-diagonal attention patterns.

5.3. Interpretation

The invariance of model behaviour under global phase rotation indicates that the learned strategy does not depend on a fixed alignment between the start of the sequence and the repeated pattern. Because phase rotation preserves the periodic structure of the sequence, the model can continue to exploit the same predictive rule used in the baseline setting. This result therefore suggests that the learned mechanism is insensitive to global phase shifts and motivates stronger structural interventions that disrupt the positional alignment between pattern repetitions.

6. INTERVENTION 3 – PREFIX ROTATION

6.1. Intervention Design

The positional ablation intervention (Section 5) showed that positional information is necessary for the model's behaviour. However, it remained unclear whether the model relies on a simple periodic shortcut tied to absolute positional phase (i.e., predicting tokens as a function of position mod P).

To test this, a prefix rotation was applied at evaluation time. A short random prefix of length $K = 3$ was inserted at the beginning of each sequence while the remainder of the sequence was shifted accordingly. This preserves the periodic structure of the repeated pattern but breaks its alignment with the original absolute positions.

6.2. Results

The prefix rotation had little effect on predictive performance. While full-sequence accuracy decreased due to the corrupted prefix region (0.66), accuracy within the induction region remained high (0.96), indicating that the model can still exploit the repeated structure of the sequence despite the phase shift.

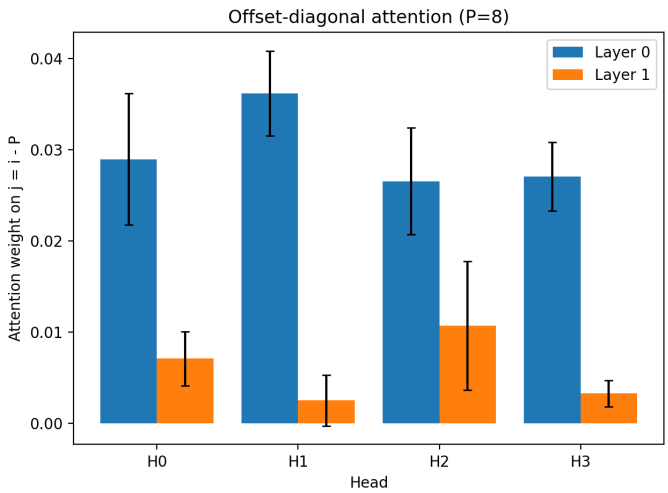

Figure 6. Mean ± standard deviation attention assigned to the offset diagonal $j = i - P$ under the prefix-rotation intervention.

Mechanistic probes reveal attention patterns similar to those observed in the baseline model. First-layer heads show only weak preference for the induction offset, while second-layer heads assign negligible attention to this diagonal (Figure 6).

6.3. Interpretation

The persistence of high induction accuracy after the prefix rotation indicates that the model does not rely on a periodic shortcut tied to absolute positional phase. Instead, the model appears able to exploit the repeated structure of the sequence in a phase-invariant manner, despite the absence of a clear induction-style attention circuit.

7. DISCUSSION AND CONCLUSION

This study examined the mechanisms learned by a small transformer trained on a synthetic periodic induction task. Although the model achieved perfect accuracy within the repeated portion of the sequence, mechanistic probes revealed no clear induction head. Attention weights showed only weak preference for the expected offset $j = i - P$, and attention maps displayed diffuse, recency-biased patterns rather than a sharp induction diagonal.

Targeted interventions further clarified the learned strategy. Removing positional embeddings caused predictive accuracy to collapse, demonstrating that the behaviour depends critically on positional information. In contrast, phase rotation and prefix perturbations did not substantially affect induction-region accuracy, indicating that the model does not rely on a fixed alignment of the repeated pattern within the sequence.

These results suggest that the model solves the task through a position-dependent shortcut rather than a canonical token-copying circuit. More broadly, the findings highlight a key interpretability lesson: strong behavioural performance does not necessarily imply that a model has learned the intended algorithmic mechanism. Even in simple synthetic settings, transformers may exploit alternative strategies that satisfy the training objective while differing substantially from the hypothesised computation.

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